

Pranav H. Premnath

Department of Physics & Astronomy
University of California, Irvine

premnatp@uci.edu

ADS · ORCID · Personal Website

Research Interests

Exoplanets • Stellar Astrophysics • Radial Velocities • Transit Photometry • Exoplanet Occurrence Rates • Bayesian Statistics

Education

University of California, Irvine 2022–Present

Ph.D. Physics (Concentration in Astronomy and Astrophysics), expected 2027

Advisor: Paul Robertson

M.S. Physics, 2024

The University of Texas at Austin 2017–2021

B.S. Astronomy

B.S. Physics

Major GPA: 3.71 Overall GPA: 3.69

Research Experience

University of California, Irvine 2022–Present

Graduate Student Researcher

Advisor: Paul Robertson

SETI Institute / Hat Creek Radio Observatory 2021–2022

Research Assistant

Advisor: Wael Farah

University of Texas at Austin 2020–2021

Undergraduate Researcher

Advisor: Caitlin Casey and Justin Spilker

University of Texas at Austin 2019–2020

Undergraduate Researcher

Advisor: Brendan Bowler and Ya-Lin Wu

Publications

First-author Publications

1. **Premnath** et al. 2026, “Searching for GEMS: Three warm Saturns and a super-Jupiter orbiting four early M-dwarfs”, *The Astronomical Journal*, Volume 169, Issue 2, id.149, 92 pp.
2. **Premnath** et al. 2020, “Evidence for a Buried AGN in an Extremely Bright Dusty Galaxy at $z = 2$ ”, *Research Notes of the AAS*, Volume 4, Issue 10, id.173.

3. **Premnath** et al. 2020, “Dynamical Masses of Young Stars Inferred from Two Transitions of CO with ALMA”, *Research Notes of the AAS*, Volume 4, Issue 7, id.100.

Co-authored Publications

1. Wright et al. 2026, “A Simple One-free-parameter Model of the Solar Chromosphere Explains Solar Chromospheric Line Variations Measured by NEID at WIYN during the 2023 October Solar Eclipse”, *Research Notes of the AAS*, Volume 10, Issue 1, id.14.
2. Beard et al. 2025, “Jitter Across 15 yr: Leveraging Precise Photometry from Kepler and TESS to Extract Exoplanets from Radial Velocity Time Series”, *The Astronomical Journal*, Volume 169, Issue 2, id.149, 92 pp.
3. Crossfield et al. 2025, “OrCAS: Origins, Compositions, and Atmospheres of Sub-Neptunes. I. Survey Definition”, *The Astronomical Journal*, Volume 169, Issue 2, id.89, 15 pp.
4. Kroft et al. 2025, “A Pair of Dynamically Interacting Sub-Neptunes Around TOI-6054”, *arXiv:2501.09095*.
5. Sheikh et al. 2024, “Correction to: Characterization of the repeating FRB 20220912A with the Allen Telescope Array”, *MNRAS*, Volume 534, Issue 3, pp.1949–1949.
6. Beard et al. 2024, “Utilizing Photometry from Multiple Sources to Mitigate Stellar Variability in Precise Radial Velocities: A Case Study of Kepler-21”, *The Astronomical Journal*, Volume 168, Issue 4, id.149, 17 pp.
7. Joshi et al. 2024, “Wideband detection of FRB 20240114A above 2 GHz with the Allen Telescope Array”, *The Astronomer’s Telegram*, No. 16599.
8. Sheikh et al. 2024, “Characterization of the repeating FRB 20220912A with the Allen Telescope Array”, *MNRAS*, Volume 527, Issue 4, pp.10425–10439.
9. Bright et al. 2023, “Precise measurements of self-absorbed rising reverse shock emission from gamma-ray burst 221009A”, *Nature Astronomy*, Volume 7, pp.986–995.
10. Sheikh et al. 2022, “Bright radio bursts from the active FRB 20220912A detected with the Allen Telescope Array”, *The Astronomer’s Telegram*, No. 15735.
11. Perez et al. 2022, “Breakthrough Listen Search for the WOW! Signal”, *Research Notes of the AAS*, Volume 6, Issue 9, id.197.

Presentations

- **Searching for GEMS: Building the Largest Volume-Limited Sample of Giant Planets around M Dwarfs**
Emerging Researchers in Exoplanet Science Symposium XI – University of Michigan, Ann Arbor (Poster, July 2026)
- **Searching for GEMS: Building the Largest Volume-Limited Sample of Giant Planets around M Dwarfs**
Sagan Summer Workshop – California Institute of Technology (Poster, July 2026)
- **Searching for GEMS: Three warm Saturns and a super-Jupiter orbiting four early M-dwarfs**
WIYN Board Meeting (Presentation, April 2026)

- **Searching for GEMS: Characterizing Giant Planet Candidates within 200pc**
NExSci ExSoCal Meeting – University of California, Los Angeles
(Poster, December 2025)
- **Probing Stellar Surfaces Using Transiting Exoplanets**
Know Thy Star, Know Thy Planet 2 – California Institute of Technology
(Poster, February 2025)
- **Pilot SETI Survey with the Upgraded Allen Telescope Array**
NExSci ExSoCal Meeting – California Institute of Technology
(Poster, December 2023)
- **Technosignature Searches with the Allen Telescope Array** *PSETI Symposium*
PSETI Symposium – The Pennsylvania State University
(Presentation (with W. Farah), June 2022)
- **Evidence for a Buried AGN in an Extremely Bright Dusty Galaxy at $z = 2$**
237th Meeting of the American Astronomical Society
(Poster, January 2021)
- **Dynamical Mass Measurement of a Young Star with ALMA**
235th Meeting of the American Astronomical Society
(Poster, January 2020)

Honors & Awards

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| • Sagan Workshop Travel Award | 2026 |
| • Sagan Workshop Travel Award | 2025 |
| • UT Austin College of Natural Sciences Research Distinction Winner | 2021 |
| • Dr. Arnold Romberg Endowed Scholarship Fund | 2020–2021 |
| • Karl G. Henize Endowed Scholarship | 2020 |
| • John W. Cox Endowment for Advanced Studies in Astronomy | 2020 |
| • Darrell W. Moffitt, Jr. Endowed Presidential Scholarship | 2019–2020 |
| • John W. Cox Endowment for Advanced Studies in Astronomy | 2019 |

Teaching Experience

Teaching Assistant, University of California, Irvine

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| • PHY 3B – Basic Physics II | Winter 2026 |
| • PHY 20A – Introduction to Astronomy | Winter 2026 |
| • PHY 7D – Classical Physics | Spring 2025 |
| • PHY 138 – Astrophysics of Galaxies | Winter 2025 |
| • PHY 3LB – Basic Physics Laboratory | Winter 2023, Summer 2023, Winter 2024, Winter 2025 |
| • PHY 3A – Basic Physics I | Fall 2024 |

- **PHY 3C** – Basic Physics III Fall 2024
- **PHY 7LC** – Classical Physics Laboratory Fall 2023
- **PHY 61C** – Introduction to Astrophysics Spring 2023, Spring 2024

Professional Service

- Administrator, Physics Graduate Caucus, UCI (2024–2025)
- Co-lead, Aspiring Allyship Group, UCI (2024–2025)
- Mentor, Physics and Astronomy Community Excellence, UCI (2023–Present)
- Treasurer, Astronomy Students Association, UT Austin (2019–2021)

Technical Skills

Programming: Python, Bash, Mathematica

Scientific Libraries: Astropy, NumPy, SciPy, Pandas, Matplotlib

Astronomy Software: CASA

Methods: Bayesian inference, MCMC, Gaussian processes, time-series analysis

References

Dr. Paul Robertson	paul.robertson@uci.edu
Dr. Shubham Kanodia	skanodia@carnegiescience.edu
Dr. Caleb Cañas	c.canas@nasa.gov
Dr. Samuel Halverson	samuel.halverson@jpl.nasa.gov